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Peripheral Blood Karyotyping

Patient name	: **Master. Rudra Singh**	**PIN**	: **DCC0000002923**
Gender/ Age	: **Male / 1 Year**	**Sample Number**	: **260154818**
Specimen	: **Peripheral blood**	**Sample collection date**	: **20-02-2026**
Referring Clinician	: **Dr. Sanjeev Agarwal**	**Sample receipt date**	: **23-02-2026**
Hospital/Clinic	: **Guru Kripa Jagriti Test Tube Center - Prayagaraj**	**Report Date**	: **17-03-2026**

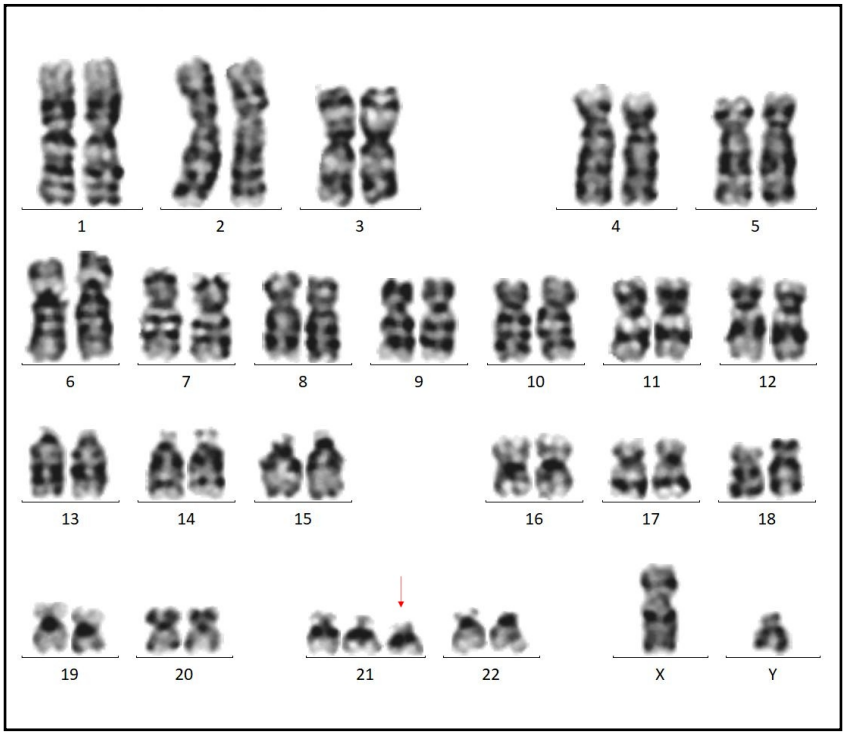
MC-6558

Test Indication

To rule out gross chromosomal abnormality

Result

International System for Human Cytogenomic Nomenclature (ISCN 2024): 47,XY,+21



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PGT-A, PGT-M & PGT-SR | Clinical Exome Sequencing | Microarray | Male Infertility | Recurrent Pregnancy Loss | Carrier Screening | Amniotic Fluid Testing | POC Analysis | NIPS
Fertility Genetics | Genetic Counseling | Oncogenetics | Neurogenetics | Infectious Genetics | New Born Screening

Metaphase analysed : 25	Autosome : Abnormal
Estimated band resolution : 475	Sex chromosome : Normal

Interpretation

The constitutional karyotype shows a male with three copies of chromosome 21, indicating Down (Trisomy 21) syndrome.

Comments

Trisomy 21 is a genetic syndrome associated with impairment of cognitive ability and physical growth as well as a particular set of facial characteristics.

Recommendations

Advised genetic counseling for the parents.

Test Methodology

A 72-hour PHA-M stimulated culture of the received peripheral blood sample was processed, according to a protocol adapted from the AGT Cytogenetics Laboratory Manual, Third Edition. Numerical and structural chromosomal abnormalities were ruled out at a banding resolution suitable for the referral indication, in accordance with current ISCN guidelines.

Limitations

- All genetic disorders cannot be ruled out by conventional karyotyping.
- The accuracy of the test is about 99%.
- G-banded analysis cannot detect small rearrangements and submicroscopic deletions.
- Low-level mosaicism may not be detected.

References

1. The AGT Cytogenetics Laboratory Manual Third Edition (1997) *Editors: Margaret J. Barch, TuridKnutsen, Jack L. Spurbeck.*
2. ISCN- An International System for Human Cytogenetic Nomenclature (2024) *Editors: Ros J. Hastings, Sarah Moore, Nicole Chia.*

This report has been reviewed and approved by:

Dr. R. Deepika., Ph.D.,	**Dr. Teena Koshy.**, Ph.D.,	**Dr. Suriya kumar G**
Consultant Cytogeneticist	Consultant Cytogeneticist	Director
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